Year 9 Engineering Technology

Engineering Technology 7–10 (2024) | 100 periods | New syllabus for 2027 | v2.0

Duration	100 × 60-minute periods	**Course**	Stage 5 / Year 9 delivery
Balance	20 theory | 20 folio | 60 practical	**Practical**	60% of course
Syllabus	Engineering Technology 7–10 (2024)	**Implementation**	From 2027
Status	Faculty-ready network program	**Local controls**	Teacher to confirm before use

Official authority: **Engineering Technology 7–10 Syllabus (2024)** | **Open the Engineering course site**

Course requirement: this 100-period program is one half of the school's 200-hour course. The master program confirms full four-focus-area coverage across both years.

Exact Stage 5 outcomes

Outcome	Exact official wording
EGT5-COM-01	communicates ideas, concepts and solutions for engineering practice
EGT5-ENV-01	analyses relationships between engineering design, production and sustainability
EGT5-EVL-01	investigates and evaluates engineering systems and solutions
EGT5-GRP-01	develops and applies technical graphics in engineering contexts
EGT5-IVT-01	explains engineering practices and the influence of technologies used in engineering industries
EGT5-MEA-01	applies mechanical analysis and practical testing to investigate engineering concepts
EGT5-SAF-01	applies risk management and safe work practices in engineering contexts
EGT5-USE-01	selects and applies materials for engineering projects

Teacher to confirm before use

ID	Confirm	Owner
TTC-ENG-01	Confirm the local timetable, project order, room allocation, class supports and approved adjustments.	Teacher / faculty
TTC-ENG-02	Confirm current assessment authority, task wording, weights, dates, conditions and submission methods.	Teacher / faculty
TTC-ENG-03	Confirm each practical step, materials, tools, machines, equipment, demonstrations, supervision, SOPs, PPE and current risk controls.	Teacher / faculty
TTC-ENG-04	Confirm collaborative testing groups, student roles, equitable participation and the evidence record.	Teacher / faculty

Teaching program | Year 9 Engineering Technology | Periods 1–5

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
1	**Site/theory**	Water Tower | Engineering principles	Teach and unpack this syllabus learning in the Water Tower context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Water Tower learning module 1** **Open the Water Tower project page**	**Describe the function and purpose of structures , including how they provide access and shelter Official content EGT5-IVT-01**	Module response showing: Describe the function and purpose of structures , including how they provide access and shelter EGT5-IVT-01
2	**Project/folio**	Water Tower | Engineering principles	Students apply the syllabus learning to their Water Tower folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Water Tower folio** **Open the Water Tower project page**	**Identify functional engineering components that provide structural integrity, such as foundations, beams and… Official content EGT5-USE-01, EGT5-COM-01**	Folio evidence demonstrating: Identify functional engineering components that provide structural integrity, such as foundations, beams and bracing EGT5-USE-01, EGT5-COM-01
3	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Outline regulations, codes of practice and standards for an identified structure in Australia Official content EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Outline regulations, codes of practice and standards for an identified structure in Australia EGT5-SAF-01, EGT5-USE-01
4	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Apply scheduling, resource allocation and budgeting to plan and manage a structural engineering project Official content EGT5-ENV-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Apply scheduling, resource allocation and budgeting to plan and manage a structural engineering project EGT5-ENV-01, EGT5-SAF-01, EGT5-USE-01
5	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	Outline the importance of Indigenous Cultural and Intellectual Property (ICIP) of Aboriginal and/or Torres… **Official content EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Outline the importance of Indigenous Cultural and Intellectual Property (ICIP) of Aboriginal and/or Torres Strait Islander Peoples and how it is related to structures EGT5-SAF-01, EGT5-USE-01

Teaching program | Year 9 Engineering Technology | Periods 6–10

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
6	**Site/theory**	Water Tower | Engineering principles	Teach and unpack this syllabus learning in the Water Tower context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Water Tower learning module 1** **Open the Water Tower project page**	**Outline the importance of Indigenous Cultural and Intellectual Property (ICIP) of Aboriginal and/or Torres… Official content** **EGT5-IVT-01**	Module response showing: Outline the importance of Indigenous Cultural and Intellectual Property (ICIP) of Aboriginal and/or Torres Strait Islander Peoples and how it is related to structures EGT5-IVT-01
7	**Project/folio**	Water Tower | Engineering principles	Students apply the syllabus learning to their Water Tower folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Water Tower folio** **Open the Water Tower project page**	**Investigate and identify the impact of a range of structures on the physical environment in Australia Official content** **EGT5-ENV-01, EGT5-COM-01**	Folio evidence demonstrating: Investigate and identify the impact of a range of structures on the physical environment in Australia EGT5-ENV-01, EGT5-COM-01
8	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Identify and apply engineering principles and processes, including types of forces and their effects… Official content** **EGT5-SAF-01, EGT5-USE-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Identify and apply engineering principles and processes, including types of forces and their effects, structural integrity and safety, to produce a functioning structure EGT5-SAF-01, EGT5-USE-01, EGT5-MEA-01
9	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	Discuss technical and enterprise skills used by structural engineers **Official content** **EGT5-IVT-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Discuss technical and enterprise skills used by structural engineers EGT5-IVT-01, EGT5-SAF-01, EGT5-USE-01
10	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Identify and implement safe work practices when producing a functioning structure Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Identify and implement safe work practices when producing a functioning structure EGT5-SAF-01, EGT5-USE-01

Teaching program | Year 9 Engineering Technology | Periods 11–15

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
11	**Site/theory**	Water Tower | Engineering principles	Teach and unpack this syllabus learning in the Water Tower context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Water Tower learning module 2** **Open the Water Tower project page**	**Discuss technical and enterprise skills used by structural engineers Official content** **EGT5-IVT-01**	Module response showing: Discuss technical and enterprise skills used by structural engineers EGT5-IVT-01
12	**Project/folio**	Water Tower | Engineering principles	Students apply the syllabus learning to their Water Tower folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Water Tower folio** **Open the Water Tower project page**	**Discuss sustainability and environmental considerations in structural or civil construction Official content** **EGT5-ENV-01, EGT5-USE-01, EGT5-COM-01**	Folio evidence demonstrating: Discuss sustainability and environmental considerations in structural or civil construction EGT5-ENV-01, EGT5-USE-01, EGT5-COM-01
13	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	Apply safe work practices throughout the design, production and testing of models and projects for a structure **Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Apply safe work practices throughout the design, production and testing of models and projects for a structure EGT5-SAF-01, EGT5-EVL-01, EGT5-USE-01
14	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Use spreadsheets to calculate material quantities and monitor project costs Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use spreadsheets to calculate material quantities and monitor project costs EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01
15	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Investigate engineered structures used by Aboriginal and/or Torres Strait Islander Peoples, such as shelters… Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Investigate engineered structures used by Aboriginal and/or Torres Strait Islander Peoples, such as shelters and fish traps EGT5-SAF-01, EGT5-USE-01

Teaching program | Year 9 Engineering Technology | Periods 16–20

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
16	**Site/theory**	Water Tower | Materials	Teach and unpack this syllabus learning in the Water Tower context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Water Tower learning module 3** **Open the Water Tower project page**	**Apply safe work practices throughout the design, production and testing of models and projects for a structure Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Apply safe work practices throughout the design, production and testing of models and projects for a structure EGT5-SAF-01, EGT5-EVL-01, EGT5-IVT-01
17	**Project/folio**	Water Tower | Materials	Students apply the syllabus learning to their Water Tower folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Water Tower folio** **Open the Water Tower project page**	Identify how materials can be modified to improve properties, such as composite materials, laminates, heat… **Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Folio evidence demonstrating: Identify how materials can be modified to improve properties, such as composite materials, laminates, heat treatment and reinforcement EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
18	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Classify engineering materials used in structures as pure metals, alloys, polymers and timbers Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Classify engineering materials used in structures as pure metals, alloys, polymers and timbers EGT5-USE-01, EGT5-SAF-01
19	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Identify the properties of a material, such as hardness, ductility, malleability, and tensile and compressive… Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Identify the properties of a material, such as hardness, ductility, malleability, and tensile and compressive strength, that make the materials suitable for use in structures EGT5-USE-01, EGT5-SAF-01
20	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Compare traditional and modern construction materials and describe their properties, such as timber versus… Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Compare traditional and modern construction materials and describe their properties, such as timber versus laminates and rammed earth versus concrete EGT5-USE-01, EGT5-SAF-01

Teaching program | Year 9 Engineering Technology | Periods 21–25

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
21	**Site/theory**	Water Tower | Engineering principles	Teach and unpack this syllabus learning in the Water Tower context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Water Tower learning module 3** **Open the Water Tower project page**	Identify functional engineering components that provide structural integrity, such as foundations, beams and… **Official content** **EGT5-USE-01, EGT5-IVT-01**	Module response showing: Identify functional engineering components that provide structural integrity, such as foundations, beams and bracing EGT5-USE-01, EGT5-IVT-01
22	**Project/folio**	Water Tower | Materials	Students apply the syllabus learning to their Water Tower folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Water Tower folio** **Open the Water Tower project page**	**Identify how materials can be modified to improve properties, such as composite materials, laminates, heat… Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Folio evidence demonstrating: Identify how materials can be modified to improve properties, such as composite materials, laminates, heat treatment and reinforcement EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
23	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Compare renewable and non-renewable resources and explain their advantages and limitations in the design and… Official content** **EGT5-ENV-01, EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Compare renewable and non-renewable resources and explain their advantages and limitations in the design and construction of structures EGT5-ENV-01, EGT5-USE-01, EGT5-SAF-01
24	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Test a range of materials to determine the properties of hardness, corrosion resistance and malleability, and… Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Test a range of materials to determine the properties of hardness, corrosion resistance and malleability, and record collected data EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01
25	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	Investigate and identify the impact of a range of structures on the physical environment in Australia **Official content** **EGT5-ENV-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Investigate and identify the impact of a range of structures on the physical environment in Australia EGT5-ENV-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Year 9 Engineering Technology | Periods 26–30

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
26	**Site/theory**	Concrete Beam | Materials	Teach and unpack this syllabus learning in the Concrete Beam context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Concrete Beam learning module 1** **Open the Concrete Beam project page**	**Select and test the suitability of materials by constructing a project** Official content **EGT5-USE-01, EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Select and test the suitability of materials by constructing a project EGT5-USE-01, EGT5-EVL-01, EGT5-IVT-01
27	**Project/folio**	Concrete Beam | Materials	Students apply the syllabus learning to their Concrete Beam folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Concrete Beam folio** **Open the Concrete Beam project page**	**Investigate engineering materials used by Aboriginal and/or Torres Strait Islander Peoples, such as timber… Official content** **EGT5-USE-01, EGT5-COM-01**	Folio evidence demonstrating: Investigate engineering materials used by Aboriginal and/or Torres Strait Islander Peoples, such as timber, resin and fibres EGT5-USE-01, EGT5-COM-01
28	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Identify forces that act on structures, including live loads , dead loads and calculate weight force** Official content **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Identify forces that act on structures, including live loads , dead loads and calculate weight force EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
29	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	Identify compressive and tensile forces acting on members in an identified structure **Official content** **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Identify compressive and tensile forces acting on members in an identified structure EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
30	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Identify compressive and tensile forces acting on members in an identified structure** Official content **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Identify compressive and tensile forces acting on members in an identified structure EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Year 9 Engineering Technology | Periods 31–35

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
31	**Site/theory**	Concrete Beam | Structural analysis	Teach and unpack this syllabus learning in the Concrete Beam context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Concrete Beam learning module 1** **Open the Concrete Beam project page**	**Construct pin-jointed structures using different shapes to explore basic structural concepts, such as force…** **Official content** **EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Construct pin-jointed structures using different shapes to explore basic structural concepts, such as force, load , reactions and equilibrium EGT5-MEA-01, EGT5-IVT-01
32	**Project/folio**	Concrete Beam | Structural analysis	Students apply the syllabus learning to their Concrete Beam folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Concrete Beam folio** **Open the Concrete Beam project page**	**Contrast destructive and non-destructive testing and how they are used to inform engineering decisions** **Official content** **EGT5-EVL-01, EGT5-COM-01**	Folio evidence demonstrating: Contrast destructive and non-destructive testing and how they are used to inform engineering decisions EGT5-EVL-01, EGT5-COM-01
33	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	Use the results of testing to modify pin-jointed structures **Official content** **EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Use the results of testing to modify pin-jointed structures EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01
34	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Test and document the effects of identified forces on pin-jointed structures** **Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Test and document the effects of identified forces on pin-jointed structures EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01
35	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Use the results of testing to modify pin-jointed structures** **Official content** **EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Use the results of testing to modify pin-jointed structures EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Year 9 Engineering Technology | Periods 36–40

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
36	**Site/theory**	Concrete Beam | Communication	Teach and unpack this syllabus learning in the Concrete Beam context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Concrete Beam learning module 2** **Open the Concrete Beam project page**	Identify key components of an engineering drawing, and the standards from AS 1100 relevant to the engineering… **Official content** **EGT5-SAF-01, EGT5-GRP-01, EGT5-IVT-01**	Module response showing: Identify key components of an engineering drawing, and the standards from AS 1100 relevant to the engineering profession, when drawing structures EGT5-SAF-01, EGT5-GRP-01, EGT5-IVT-01
37	**Project/folio**	Concrete Beam | Structural analysis	Students apply the syllabus learning to their Concrete Beam folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Concrete Beam folio** **Open the Concrete Beam project page**	**Build and test a working model of a structure for a specific purpose using appropriate tools and processes Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-COM-01**	Folio evidence demonstrating: Build and test a working model of a structure for a specific purpose using appropriate tools and processes EGT5-USE-01, EGT5-EVL-01, EGT5-COM-01
38	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Evaluate the use of structural elements in projects Official content** **EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Evaluate the use of structural elements in projects EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01
39	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Identify key components of an engineering drawing, and the standards from AS 1100 relevant to the engineering… Official content** **EGT5-SAF-01, EGT5-GRP-01, EGT5-IVT-01**	Teacher observation and project/test record demonstrating: Identify key components of an engineering drawing, and the standards from AS 1100 relevant to the engineering profession, when drawing structures EGT5-SAF-01, EGT5-GRP-01, EGT5-IVT-01
40	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	Document material selection and justification, structural analysis and test results in an engineering report **Official content** **EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01**	Teacher observation and project/test record demonstrating: Document material selection and justification, structural analysis and test results in an engineering report EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01

Teaching program | Year 9 Engineering Technology | Periods 41–45

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
41	**Site/theory**	Concrete Beam | Communication	Teach and unpack this syllabus learning in the Concrete Beam context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Concrete Beam learning module 3** **Open the Concrete Beam project page**	**Produce a pictorial sketch of an identified structure, such as an isometric or oblique drawing** **Official content** **EGT5-GRP-01, EGT5-USE-01, EGT5-IVT-01**	Module response showing: Produce a pictorial sketch of an identified structure, such as an isometric or oblique drawing EGT5-GRP-01, EGT5-USE-01, EGT5-IVT-01
42	**Project/folio**	Concrete Beam | Communication	Students apply the syllabus learning to their Concrete Beam folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Concrete Beam folio** **Open the Concrete Beam project page**	**Develop an orthogonal engineering drawing of a structure, including 2 views and dimensions using third-angle…** **Official content** **EGT5-GRP-01, EGT5-COM-01**	Folio evidence demonstrating: Develop an orthogonal engineering drawing of a structure, including 2 views and dimensions using third-angle projection EGT5-GRP-01, EGT5-COM-01
43	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	Use subject-specific terminology to communicate concepts of engineering structures **Official content** **EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Use subject-specific terminology to communicate concepts of engineering structures EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01
44	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Produce a parts list to prepare materials for the construction of a structure** **Official content** **EGT5-GRP-01, EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Produce a parts list to prepare materials for the construction of a structure EGT5-GRP-01, EGT5-USE-01, EGT5-SAF-01
45	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Document material selection and justification, structural analysis and test results in an engineering report** **Official content** **EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01**	Teacher observation and project/test record demonstrating: Document material selection and justification, structural analysis and test results in an engineering report EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01

Teaching program | Year 9 Engineering Technology | Periods 46–50

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
46	**Site/theory**	Concrete Beam | Communication	Teach and unpack this syllabus learning in the Concrete Beam context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Concrete Beam learning module 3** **Open the Concrete Beam project page**	**Apply data presentation techniques when presenting structural test results Official content** **EGT5-COM-01, EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Apply data presentation techniques when presenting structural test results EGT5-COM-01, EGT5-EVL-01, EGT5-IVT-01
47	**Project/folio**	Concrete Beam | Structural analysis	Students apply the syllabus learning to their Concrete Beam folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Concrete Beam folio** **Open the Concrete Beam project page**	Identify compressive and tensile forces acting on members in an identified structure **Official content** **EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Identify compressive and tensile forces acting on members in an identified structure EGT5-MEA-01, EGT5-COM-01
48	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Create written texts to explain and evaluate factors that influence the design and engineering of structures Official content** **EGT5-COM-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Create written texts to explain and evaluate factors that influence the design and engineering of structures EGT5-COM-01, EGT5-EVL-01, EGT5-SAF-01
49	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Use subject-specific terminology to communicate concepts of engineering structures Official content** **EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Use subject-specific terminology to communicate concepts of engineering structures EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01
50	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	Test and document the effects of identified forces on pin-jointed structures **Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Test and document the effects of identified forces on pin-jointed structures EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Year 9 Engineering Technology | Periods 51–55

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
51	**Site/theory**	Hydraulic Digger | Engineering principles	Teach and unpack this syllabus learning in the Hydraulic Digger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Hydraulic Digger learning module 1** **Open the Hydraulic Digger project page**	**Describe the function and purpose of mechanisms , including how they operate as part of a system or machine** **Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Describe the function and purpose of mechanisms , including how they operate as part of a system or machine EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01
52	**Project/folio**	Hydraulic Digger | Engineering principles	Students apply the syllabus learning to their Hydraulic Digger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Hydraulic Digger folio** **Open the Hydraulic Digger project page**	**Identify the functional components of a mechanism, such as axles, levers, pulleys, springs, gears and screws** **Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Identify the functional components of a mechanism, such as axles, levers, pulleys, springs, gears and screws EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01
53	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Identify and use safe work practices when participating in testing and developing mechanical projects** **Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Identify and use safe work practices when participating in testing and developing mechanical projects EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01
54	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Dismantle and reassemble a mechanism to outline its function** **Official content** **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Dismantle and reassemble a mechanism to outline its function EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
55	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Apply scheduling, resource allocation and budgeting to plan and manage a mechanical engineering project** **Official content** **EGT5-ENV-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Apply scheduling, resource allocation and budgeting to plan and manage a mechanical engineering project EGT5-ENV-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Year 9 Engineering Technology | Periods 56–60

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
56	**Site/theory**	Hydraulic Digger | Engineering principles	Teach and unpack this syllabus learning in the Hydraulic Digger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Hydraulic Digger learning module 1** **Open the Hydraulic Digger project page**	**Investigate how Aboriginal and/or Torres Strait Islander Peoples used levers and pulleys Official content EGT5-IVT-01**	Module response showing: Investigate how Aboriginal and/or Torres Strait Islander Peoples used levers and pulleys EGT5-IVT-01
57	**Project/folio**	Hydraulic Digger | Engineering principles	Students apply the syllabus learning to their Hydraulic Digger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Hydraulic Digger folio** **Open the Hydraulic Digger project page**	**Test the mechanical advantage and efficiency of simple machines , including a lever, wheel and axle, pulley… Official content EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Folio evidence demonstrating: Test the mechanical advantage and efficiency of simple machines , including a lever, wheel and axle, pulley, gear and spring EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
58	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Describe key innovations based on simple machines to understand how mechanical engineering has evolved over time Official content EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Describe key innovations based on simple machines to understand how mechanical engineering has evolved over time EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01
59	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	Explain how simple machines work together in a mechanical system **Official content EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Explain how simple machines work together in a mechanical system EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01
60	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Explain how simple machines work together in a mechanical system Official content EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Explain how simple machines work together in a mechanical system EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Year 9 Engineering Technology | Periods 61–65

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
61	**Site/theory**	Hydraulic Digger | Engineering principles	Teach and unpack this syllabus learning in the Hydraulic Digger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Hydraulic Digger learning module 2** **Open the Hydraulic Digger project page**	**Identify and apply linear, rotary, oscillating and reciprocating motion in various mechanical devices Official content** **EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Identify and apply linear, rotary, oscillating and reciprocating motion in various mechanical devices EGT5-MEA-01, EGT5-IVT-01
62	**Project/folio**	Hydraulic Digger | Engineering principles	Students apply the syllabus learning to their Hydraulic Digger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Hydraulic Digger folio** **Open the Hydraulic Digger project page**	**Use and modify existing designs when completing projects Official content** **EGT5-EVL-01, EGT5-COM-01**	Folio evidence demonstrating: Use and modify existing designs when completing projects EGT5-EVL-01, EGT5-COM-01
63	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Construct a model of a mechanism that converts circular motion to linear motion Official content** **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Construct a model of a mechanism that converts circular motion to linear motion EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
64	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Discuss the application of precision measuring tools, such as micrometers and vernier calipers, to ensure… Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Discuss the application of precision measuring tools, such as micrometers and vernier calipers, to ensure accuracy in the production of mechanical parts EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01
65	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Calculate quantities and costs of materials and components used in the completion of mechanical projects Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Calculate quantities and costs of materials and components used in the completion of mechanical projects EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Year 9 Engineering Technology | Periods 66–70

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
66	**Site/theory**	Hydraulic Digger | Engineering principles	Teach and unpack this syllabus learning in the Hydraulic Digger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Hydraulic Digger learning module 3** **Open the Hydraulic Digger project page**	**Investigate the role of mechanical engineering in the development of robotics used in automation Official content** **EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Investigate the role of mechanical engineering in the development of robotics used in automation EGT5-MEA-01, EGT5-IVT-01
67	**Project/folio**	Hydraulic Digger | Engineering principles	Students apply the syllabus learning to their Hydraulic Digger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Hydraulic Digger folio** **Open the Hydraulic Digger project page**	Evaluate and use mechanical components to solve an identified problem in practical projects **Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Folio evidence demonstrating: Evaluate and use mechanical components to solve an identified problem in practical projects EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
68	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Discuss sustainability considerations in mechanical engineering, such as product life cycle and reusability of… Official content** **EGT5-ENV-01, EGT5-USE-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Discuss sustainability considerations in mechanical engineering, such as product life cycle and reusability of machine parts EGT5-ENV-01, EGT5-USE-01, EGT5-MEA-01
69	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Evaluate and use mechanical components to solve an identified problem in practical projects Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Evaluate and use mechanical components to solve an identified problem in practical projects EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
70	**Practical**	Hydraulic Digger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Apply safe work practices throughout the design, production and testing of models and projects for a mechanism Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Apply safe work practices throughout the design, production and testing of models and projects for a mechanism EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01

Teaching program | Year 9 Engineering Technology | Periods 71–75

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
71	**Site/theory**	Hydraulic Digger | Materials	Teach and unpack this syllabus learning in the Hydraulic Digger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Hydraulic Digger learning module 3** **Open the Hydraulic Digger project page**	**Identify the properties of a material, including hardness, machineability and wear resistance, that make it… Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Identify the properties of a material, including hardness, machineability and wear resistance, that make it suitable for use in mechanisms EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01
72	**Project/folio**	Hydraulic Digger | Materials	Students apply the syllabus learning to their Hydraulic Digger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Hydraulic Digger folio** **Open the Hydraulic Digger project page**	**Compare the general properties of traditional and modern materials used in manufacturing mechanisms, including… Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Compare the general properties of traditional and modern materials used in manufacturing mechanisms, including metals and composites EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01
73	**Practical**	Hydraulic Digger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Apply work hardening and heat treatment processes to modify a range of materials, and test how these processes… Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Apply work hardening and heat treatment processes to modify a range of materials, and test how these processes change the properties of the material EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01
74	**Practical**	Hydraulic Digger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Use data to assess the properties of a range of materials to evaluate their performance under identified… Official content** **EGT5-ENV-01, EGT5-USE-01, EGT5-EVL-01**	Teacher observation and project/test record demonstrating: Use data to assess the properties of a range of materials to evaluate their performance under identified conditions, such as temperature extremes and corrosive environments EGT5-ENV-01, EGT5-USE-01, EGT5-EVL-01
75	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	Identify and use safe work practices when participating in testing and developing mechanical projects **Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Identify and use safe work practices when participating in testing and developing mechanical projects EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01

Teaching program | Year 9 Engineering Technology | Periods 76–80

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
76	**Site/theory**	Geared Lolly Dispenser | Materials	Teach and unpack this syllabus learning in the Geared Lolly Dispenser context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Geared Lolly Dispenser learning module 1** **Open the Geared Lolly Dispenser project page**	**Select and assess the suitability of materials used to construct a mechanism for an intended purpose Official content EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Select and assess the suitability of materials used to construct a mechanism for an intended purpose EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01
77	**Project/folio**	Geared Lolly Dispenser | Materials	Students apply the syllabus learning to their Geared Lolly Dispenser folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Geared Lolly Dispenser evidence guide** **Open the Geared Lolly Dispenser project page**	**Investigate the effect of friction between materials, including how friction could be either an advantage or… Official content EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Investigate the effect of friction between materials, including how friction could be either an advantage or disadvantage in a mechanism EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01
78	**Practical**	Geared Lolly Dispenser | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Test the factors that impact on the corrosion of a mechanical component Official content EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Test the factors that impact on the corrosion of a mechanical component EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
79	**Practical**	Geared Lolly Dispenser | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Outline and apply a range of processes and techniques, such as the application of finishes, to protect… Official content EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Outline and apply a range of processes and techniques, such as the application of finishes, to protect mechanical components from corrosion EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01
80	**Practical**	Geared Lolly Dispenser | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Investigate the role of the waste hierarchy in material selection and use during the production of mechanisms Official content EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Investigate the role of the waste hierarchy in material selection and use during the production of mechanisms EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Year 9 Engineering Technology | Periods 81–85

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
81	**Site/theory**	Geared Lolly Dispenser | Mechanical analysis	Teach and unpack this syllabus learning in the Geared Lolly Dispenser context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Geared Lolly Dispenser learning module 1** **Open the Geared Lolly Dispenser project page**	**Test a range of simple mechanisms, including levers, pulleys and gears, to determine their efficiency** Official content **EGT5-EVL-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Test a range of simple mechanisms, including levers, pulleys and gears, to determine their efficiency EGT5-EVL-01, EGT5-MEA-01, EGT5-IVT-01
82	**Project/folio**	Geared Lolly Dispenser | Mechanical analysis	Students apply the syllabus learning to their Geared Lolly Dispenser folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Geared Lolly Dispenser evidence guide** **Open the Geared Lolly Dispenser project page**	Analyse the functional effect of force and the related output of simple mechanisms, such as levers, pulleys… **Official content** **EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Analyse the functional effect of force and the related output of simple mechanisms, such as levers, pulleys and gears EGT5-MEA-01, EGT5-COM-01
83	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Analyse the functional effect of force and the related output of simple mechanisms, such as levers, pulleys…** Official content **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Analyse the functional effect of force and the related output of simple mechanisms, such as levers, pulleys and gears EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
84	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Build and test a working model of a mechanism using appropriate tools and processes** Official content **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Build and test a working model of a mechanism using appropriate tools and processes EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
85	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Compare the forces acting on a model of a mechanism and assess their efficiency** Official content **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Compare the forces acting on a model of a mechanism and assess their efficiency EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Year 9 Engineering Technology | Periods 86–90

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
86	**Site/theory**	Geared Lolly Dispenser | Mechanical analysis	Teach and unpack this syllabus learning in the Geared Lolly Dispenser context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Geared Lolly Dispenser learning module 2** **Open the Geared Lolly Dispenser project page**	**Test and evaluate how mechanical systems relate to and interact with other systems to demonstrate their… Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Test and evaluate how mechanical systems relate to and interact with other systems to demonstrate their dependencies on each other EGT5-EVL-01, EGT5-MEA-01, EGT5-IVT-01
87	**Project/folio**	Geared Lolly Dispenser | Mechanical analysis	Students apply the syllabus learning to their Geared Lolly Dispenser folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Geared Lolly Dispenser evidence guide** **Open the Geared Lolly Dispenser project page**	**Apply an engineering process to produce a functioning mechanism based on mechanical principles Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Apply an engineering process to produce a functioning mechanism based on mechanical principles EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01
88	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	Evaluate the use of mechanical elements in projects **Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Evaluate the use of mechanical elements in projects EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01
89	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Identify and explain the concept of mechanical advantage (MA) in a simple mechanical system Official content** **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Identify and explain the concept of mechanical advantage (MA) in a simple mechanical system EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
90	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Evaluate the use of mechanical elements in projects Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Evaluate the use of mechanical elements in projects EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Year 9 Engineering Technology | Periods 91–95

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
91	**Site/theory**	Geared Lolly Dispenser | Communication	Teach and unpack this syllabus learning in the Geared Lolly Dispenser context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Geared Lolly Dispenser learning module 3** **Open the Geared Lolly Dispenser project page**	**Develop an engineering drawing using the AS 1100 for drawing mechanisms** **Official content** **EGT5-GRP-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Develop an engineering drawing using the AS 1100 for drawing mechanisms EGT5-GRP-01, EGT5-MEA-01, EGT5-IVT-01
92	**Project/folio**	Geared Lolly Dispenser | Communication	Students apply the syllabus learning to their Geared Lolly Dispenser folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Geared Lolly Dispenser evidence guide** **Open the Geared Lolly Dispenser project page**	**Produce a pictorial drawing of a simple mechanical component, such as an isometric or oblique drawing** **Official content** **EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01**	Folio evidence demonstrating: Produce a pictorial drawing of a simple mechanical component, such as an isometric or oblique drawing EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01
93	**Practical**	Geared Lolly Dispenser | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Develop an orthogonal engineering drawing of a mechanical component, including 2 views and dimensions using…** **Official content** **EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Develop an orthogonal engineering drawing of a mechanical component, including 2 views and dimensions using third-angle projection EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01
94	**Practical**	Geared Lolly Dispenser | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	Document material selection and justification, mechanical analysis and test results in an engineering report **Official content** **EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01**	Teacher observation and project/test record demonstrating: Document material selection and justification, mechanical analysis and test results in an engineering report EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01
95	**Practical**	Geared Lolly Dispenser | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Apply a sequence of skills to develop an engineering drawing using computer-aided design (CAD) software** **Official content** **EGT5-GRP-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Apply a sequence of skills to develop an engineering drawing using computer-aided design (CAD) software EGT5-GRP-01, EGT5-SAF-01, EGT5-USE-01

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P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
96	Site/theory	Geared Lolly Dispenser | Communication	Teach and unpack this syllabus learning in the Geared Lolly Dispenser context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Geared Lolly Dispenser learning module 3** **Open the Geared Lolly Dispenser project page**	**Produce a parts list to prepare materials for the construction of a mechanism** Official content **EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01**	Module response showing: Produce a parts list to prepare materials for the construction of a mechanism EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01
97	Project/folio	Geared Lolly Dispenser | Communication	Students apply the syllabus learning to their Geared Lolly Dispenser folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Geared Lolly Dispenser evidence guide** **Open the Geared Lolly Dispenser project page**	**Document material selection and justification, mechanical analysis and test results in an engineering report** Official content **EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01**	Folio evidence demonstrating: Document material selection and justification, mechanical analysis and test results in an engineering report EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01
98	Practical	Geared Lolly Dispenser | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Create written texts to explain and justify decision-making in the development of a solution** Official content **EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Create written texts to explain and justify decision-making in the development of a solution EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01
99	Practical	Geared Lolly Dispenser | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Use subject-specific terminology to communicate concepts of engineering mechanisms** Official content **EGT5-COM-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use subject-specific terminology to communicate concepts of engineering mechanisms EGT5-COM-01, EGT5-MEA-01, EGT5-SAF-01
100	Practical	Geared Lolly Dispenser | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	Outline and apply a range of processes and techniques, such as the application of finishes, to protect… Official content **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Outline and apply a range of processes and techniques, such as the application of finishes, to protect mechanical components from corrosion EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01